

Metabase Dashboard Project with PostgreSQL (Big Data)

It's time for my capstone project!

This final assignment will test my SQL skills, Metabase dashboarding expertise, and your ability to generate insights from **real-world retail business data**.

I will be working with the **Mexico Toy Sales & Inventory Dataset**, which contains products, stores, daily sales, and stock levels for a fictional toy store chain operating across Mexico.

Let's turn my learning into a portfolio-ready BI project!

Dataset Description:

You are provided with **four interconnected tables**:

- **Products**
- **Stores**
- **Sales**
- **Inventory**

Business questions:

Profitability

- Which categories drive the most profit?
- Which products have the highest margin?
- Which stores generate the most revenue?

Store Performance

- Which cities sell the most units?
- Which stores underperform?

Seasonality & Trends

- Monthly or weekly sales trends
- Products with seasonal spikes

Inventory & Stock-Out Risk

- Any store running out of high-demand SKUs?
- Total value of inventory tied up
- Products needing urgent restocking

High-Impact Insights & Recommended Actions

1. Critical Inventory Failures on Top Sellers The most alarming metric is the inventory Status. There are 1,089 rows of products flagged as "Needs Restock".

Zero Stock on Volume Drivers: High-volume items like "Action Figures" (57,958 sold) and "Dino Eggs" (38,236 sold) currently have 0 stock in hand .

Zero Stock on Brand Names: Iconic items like Jenga, Etch A Sketch, and Hot Wheels 5-Packs are completely stocked out.

Impact: You are losing sales daily on products that have proven demand.

2. **Category Strategy:** Volume vs. Efficiency There is a distinct split between what drives *volume* versus what drives *efficiency* :

Volume Driver: The "Toys" category is the largest contributor to total profit (29.5%) .

Margin Driver: "Electronics" have the highest profit margin per unit at 44.57%, significantly higher than Toys (21.2%).

Implication: While Toys keep the lights on, Electronics generate more profit for every dollar of revenue earned.

3. **Geographic Concentration & Underperformance**

Top Performer: Sales are heavily concentrated in Mexico City (Ciudad de Mexico), which drives the highest unit sales by a large margin (125.6k units) compared to the next runner-up (Guadalajara at 96.5k).

Revenue Disparity: The top store (Ciudad de Mexico 2) generates \$554.5k, while underperforming stores (like Saltillo 1 and Monterrey 4) are generating significantly less revenue (dashboard indicates a steep drop-off in the bar charts) .

4. **Seasonality Risks**

Specific products like Monopoly and Ping Pong Sets show clear seasonal spikes, particularly around December. If the current inventory issues persist into the holiday season, the missed opportunity cost will be exponential.

Recommended Actions

1. **Immediate Operational Fix: Emergency Restocking**

Action: Immediately place expedited purchase orders for the "Zero Stock" items identified in the "Products needing urgent restocking" table.

Priority: Prioritize Action Figures and Dino Eggs first, as they have the highest historical sales volume (57k+ units).

2. **Strategic Marketing Shift**

Action: Allocate a higher percentage of the marketing budget toward Electronics.

Why: With a margin of 44.57% (vs 21.2% for toys), increasing sales velocity in Electronics will have a disproportionately positive impact on the bottom line compared to selling more traditional toys.

3. **Location-Based Investigation**

Action: Audit the Saltillo 1 and Monterrey 4 locations.

Why: Determine if their underperformance is due to:

1. Stock allocations: Are they suffering from the same stockouts as the main warehouse?
2. Location: Is foot traffic too low to sustain operations? *If stock is available elsewhere, consider an inter-store transfer to Mexico City locations where demand is proven.*

4. **Seasonal Planning (Q3/Q4 Prep)**

Action: Review the "Products with seasonal spikes" list. Ensure that inventory orders for Monopoly and Chutes & Ladders are placed 3 months in advance of December to capture the holiday surge shown in the data.

Mexico Toy Sales & Inventory_dashboard

Date_Filter

Date: Month

Store Filter

Categories Filter

Store Locations

1.1M

Total Orders

14.4M

Total Revenue

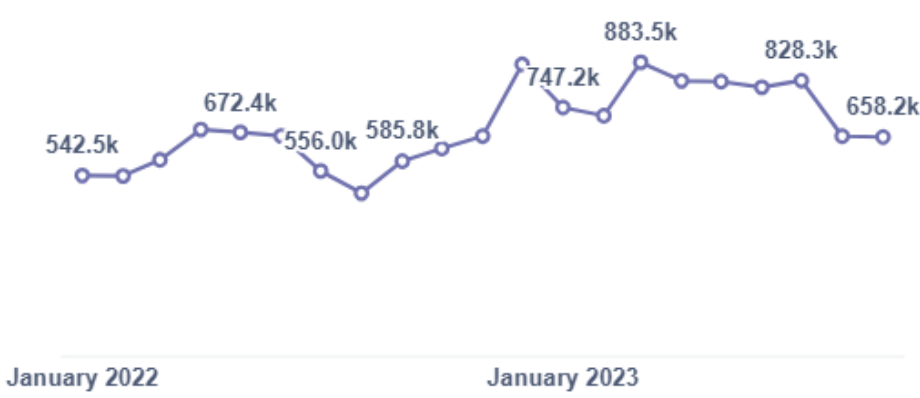
27.77%

Total profit margin

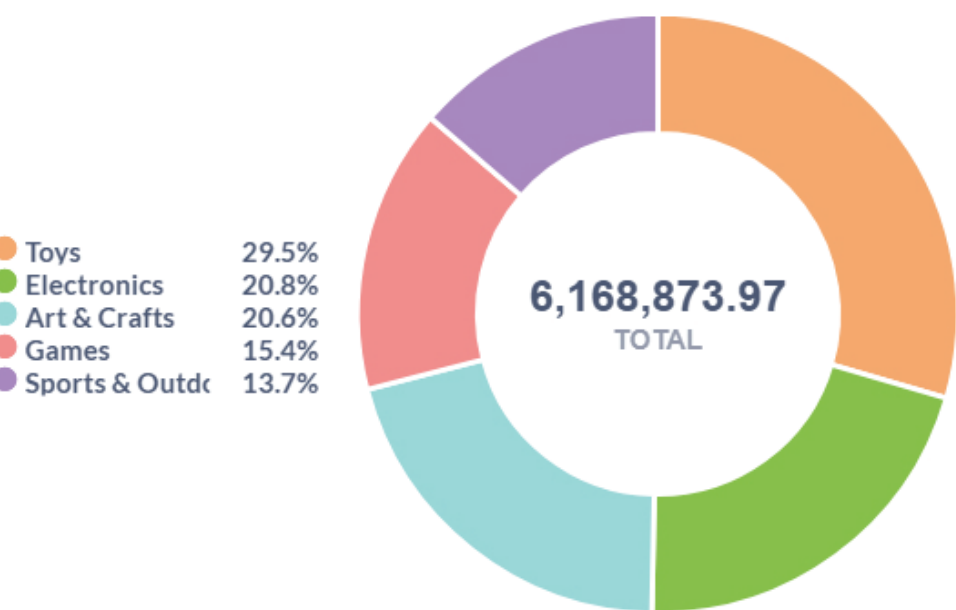
6.2M

Total Profit

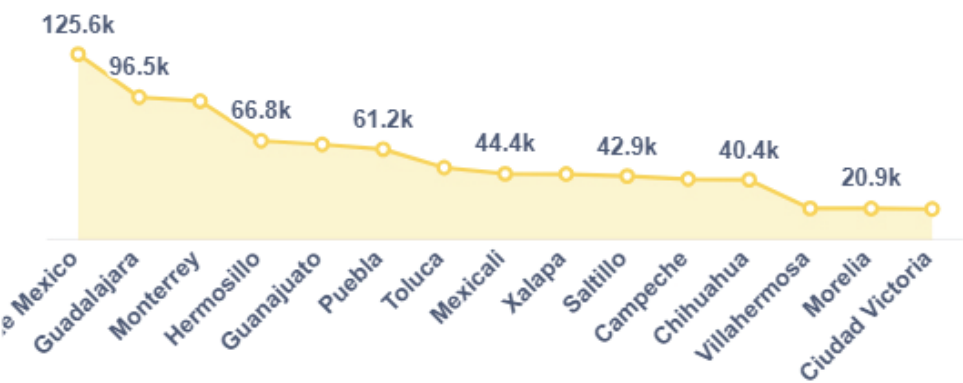
Monthly or weekly sales trends



Which categories drive the most profit?



Which cities sell the most units?



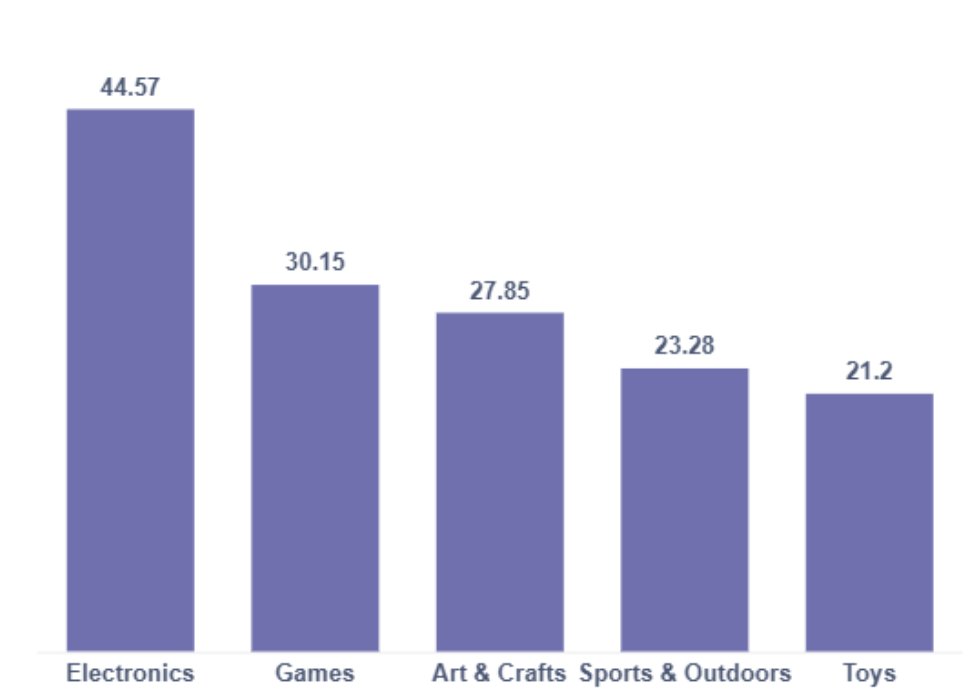
Any store running out of high-demand SKUs?

Product_ID	Product_Name	total_units_sold	stock_in_hand
21	Mini Ping Pong Set	33,604	0
1	Action Figure	57,958	0
9	Dino Egg	38,236	0
2	Animal Figures	39,089	0
9	Dino Egg	38,236	0
3	Barrel O' Slime	91,663	0
14	Glass Marbles	37,518	0
31	Splash Balls	60,248	0
378 rows			

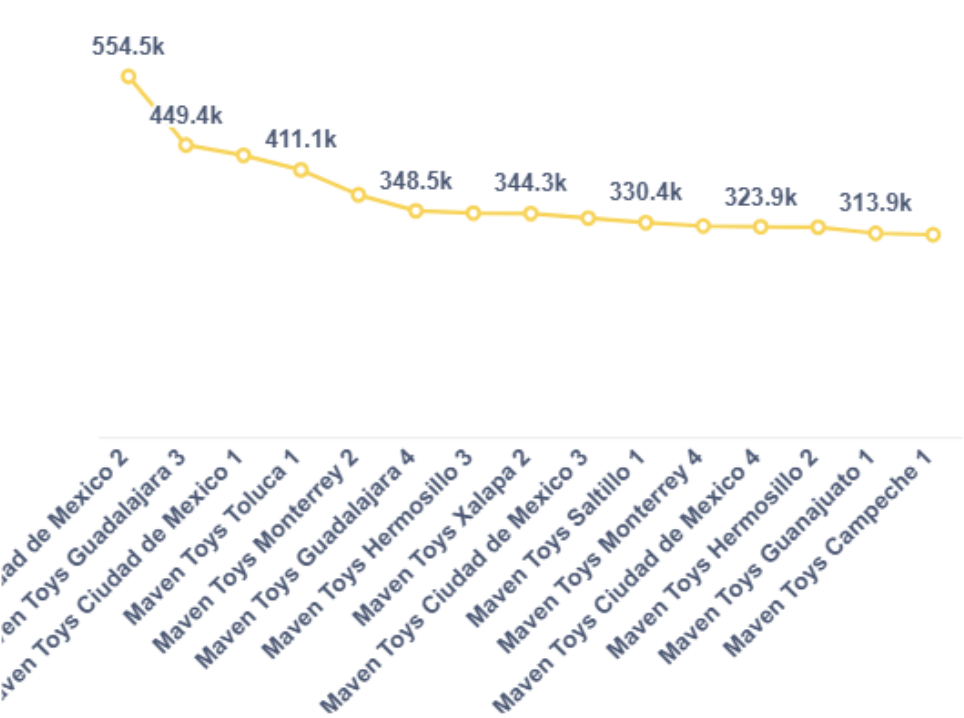
Products needing urgent restocking?

Product_ID	Product_Name	stock_in_hand	status
11	Etch A Sketch	0	⚠ Needs Restock
28	Playfoam	0	⚠ Needs Restock
1	Action Figure	0	⚠ Needs Restock
16	Jenga	0	⚠ Needs Restock
9	Dino Egg	0	⚠ Needs Restock
15	Hot Wheels 5-Pack	0	⚠ Needs Restock
13	Gamer Headphones	0	⚠ Needs Restock
14	Glass Marbles	0	⚠ Needs Restock
1,089 rows			

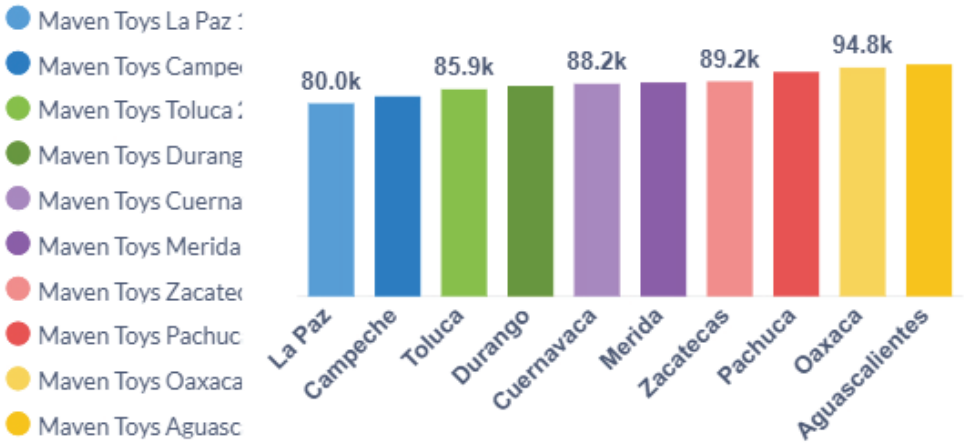
Which products have the highest margin?



Which stores generate the most revenue?



Which stores underperform?



Products with seasonal spikes

Product_Name	month	total_units	avg units
Monopoly	June 1, 2022, 12:00 AM	453	161.19
Monopoly	December 1, 2022, 12:00 AM	434	161.19
Chutes & Ladders	December 1, 2022, 12:00 AM	470	182.33
Mini Ping Pong Set	December 1, 2022, 12:00 AM	4,000	1,600.19
Splash Balls	July 1, 2022, 12:00 AM	4,702	2,848.05
132 rows			

Total value of inventory tied up!

Product_Name	Product_Inventory_Value
Lego Bricks	44,708.82
Dinosaur Figures	34,057.28
Magic Sand	30,732.78
Rubik's Cube	29,925.03
Nerf Gun	24,227.88
Deck Of Cards	19,138.62
Colorbuds	17,373.41
Toy Robot	16,165.78
35 rows	